

Signals & Systems Studio

Detailed Help Book - interface, functions, syntax, transformations, analysis and examples

What this guide covers

This guide documents the current version of the Signals & Systems Studio. It explains the direct expression-entry workflow, all signal functions in the current program, the toolbar and menus, plotting controls, analysis tools, file export, keyboard shortcuts, and common signal-building patterns.

Core workflow: type a signal -> press Enter or Plot -> inspect the graph -> change the expression -> plot again.

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1. Getting started

1.1 Launching the program

Open Terminal in the project folder, activate the virtual environment, and run the program:

```
source .venv/bin/activate
python just_trying_to_study.py
```

The program requires NumPy, Matplotlib, and a Python build with Tkinter. On the Homebrew Python setup used for this project, Tkinter is supplied separately through the Homebrew Python Tk package.

1.2 Your first signal

The expression field is the most important part of the application. Click it and type:

```
r(t) - 2*r(t-1) + r(t-2)
```

Press Enter. The application evaluates the expression over the selected time range and plots the result. You can then replace the expression with another signal without restarting the program.

1.3 Expressions worth trying immediately

```
u(t)
r(t)
u(t)-u(t-3)
r(t)-r(t-2)
r(t)-2*r(t-1)+r(t-2)
delta(t-2)
r(t)+2*u(t-1)+3*delta(t-3)
```

2. Interface tour

2.1 Window structure

The current design has a fixed toolbar at the top, a narrow scrollable control sidebar on the left, and a responsive graph area on the right. The graph uses the space available to it rather than requiring a fixed application width.

Area	Purpose
Menu bar	Full access to File, Functions, Analysis, View and Help.
Toolbar	Fast access to the most common study and export actions.
Signal sidebar	Expression input, function insertion, time range, view options, constructions and analysis.
Graph	Main waveform display.
Current Signal	Shows the expression, sample count, time range, minimum and maximum.
Status area	Reports what the application just did.

2.2 Small-screen behaviour

The left sidebar is independently vertically scrollable. On macOS, trackpad wheel events are routed to the sidebar so two-finger scrolling can be used when the sidebar content is taller than the available window. The sidebar content is constrained to the visible width to avoid unnecessary horizontal spill.

2.3 Expression entry

The expression box accepts normal mathematical-style signal expressions. Function buttons insert function text at the cursor rather than forcing you to rebuild the entire expression. This is particularly useful when constructing signals from several shifted steps or ramps.

3. Toolbar

The toolbar is deliberately focused on the operations you are most likely to use while studying.

Button	Action	Typical use
Plot	Evaluate and redraw the current expression.	After editing an expression.
New	Clear the current expression and plot.	Start a new exercise.
$u(t)$	Load unit step.	Study step behaviour.
$r(t)$	Load unit ramp.	Study ramp behaviour.
$\delta(t)$	Load unit impulse.	Study impulse notation.
Pulse	Load $u(t)-u(t-3)$.	Study pulse construction.
Triangle	Load $r(t)-2*r(t-1)+r(t-2)$.	Study ramp construction.
Auto Y	Toggle Y-axis autoscaling.	Fit amplitude automatically.
Fit	Return to the default time view and autoscale.	Recover a clean view.
Zoom +	Narrow the time interval.	Inspect a waveform closely.
Zoom -	Widen the time interval.	See more of the signal.
PNG	Save plot as an image.	Notes, assignments, reports.
PDF	Save plot as vector PDF.	Printing and reports.
Theme	Toggle the light/dark theme.	Visual preference.
Help	Open the Help Book PDF.	Reference while studying.

4. File menu

Command	What it does
New / Clear	Clears the current expression and resets the plot area.
Open Expression...	Loads a signal expression from a text file.
Save Expression...	Saves the current expression to a text file.
Save as Image...	Exports the current graph as PNG, JPG or SVG.
Save as PDF...	Exports the current graph as PDF.
Export Samples as CSV...	Exports the sampled t and $x(t)$ values.
Exit	Closes the application.

4.1 Recommended workflow for assignments

Plot the required signal, choose a useful time range, enable autoscaling if needed, and use **File -> Save as Image...** for a figure. Use **Export Samples as CSV...** when numerical values are required.

5. Functions menu

The Functions menu is the main signal library. Selecting an entry puts the expression into the expression field and plots it immediately.

5.1 Basic Signals

Name	Syntax	Description
Unit step	$u(t)$	0 for $t < 0$, 1 for $t \geq 0$.
Unit ramp	$r(t)$	$\max(t, 0)$.
Impulse	$\delta(t)$	Dirac impulse, drawn as an arrow.
Impulse alias	$d(t)$	Short alias for $\delta(t)$.
Sign	$\text{sgn}(t)$	-1 for negative t , 0 at zero, +1 for positive t .
Rectangle	$\text{rect}(t)$	Unit rectangular pulse centred at zero.
Triangle	$\text{tri}(t)$	Unit triangular pulse centred at zero.
Sinc	$\text{sinc}(t)$	Normalized sinc $\sin(\pi t)/(\pi t)$.

5.2 Transformations

Preset	Expression	What it demonstrates
Delayed step	$u(t-2)$	Right shift by 2.
Advanced step	$u(t+2)$	Left shift by 2.
Delayed ramp	$r(t-2)$	Right shift by 2.
Advanced ramp	$r(t+2)$	Left shift by 2.
Compressed ramp	$r(2t)$	Time compression.
Expanded ramp	$r(0.5t)$	Time expansion.
Reversed ramp	$r(-t)$	Time reversal.
Scaled step	$3u(t)$	Amplitude scaling.
Scaled ramp	$2r(t)$	Amplitude scaling.
Scaled impulse	$4\delta(t)$	Impulse amplitude scaling.

5.3 Signal constructions

Preset	Expression
Rectangular pulse	$u(t) - u(t-3)$
Delayed pulse	$u(t-2) - u(t-5)$
Finite ramp	$r(t) - r(t-2)$
Triangle from ramps	$r(t) - 2r(t-1) + r(t-2)$
Wider triangle	$r(t) - 2r(t-2) + r(t-4)$
Step staircase	$u(t) + u(t-2) + u(t-4)$
Impulse pair	$\delta(t-2) + \delta(t+2)$

Preset	Expression
Odd impulse pair	$\delta(t+2) - \delta(t-2)$
Causal sine	$\sin(2t)u(t)$
Causal cosine	$\cos(2t)u(t)$
Damped sine	$\exp(-0.2t)\sin(2t)u(t)$
Damped cosine	$\exp(-0.2t)\cos(2t)u(t)$

6. Mathematical functions

Function	Syntax	Example
Sine	$\sin(x)$	$\sin(2*t)$
Cosine	$\cos(x)$	$\cos(3*t)$
Tangent	$\tan(x)$	$\tan(t)$
Exponential	$\exp(x)$	$\exp(-t)$
Square root	$\text{sqrt}(x)$	$\text{sqrt}(\text{abs}(t))$
Absolute value	$\text{abs}(x)$	$\text{abs}(t-2)$
Natural log	$\ln(x)$	$\ln(t+3)$
Log base 10	$\log_{10}(x)$	$\log_{10}(t+3)$
Constant pi	pi	$\sin(\text{pi}*t)$
Euler constant	e	e^{**t}

The plotting engine is numerical. Functions that would ordinarily be undefined for part of the real line are protected with numerical handling so that the graph can still render a usable result.

7. Expression syntax

7.1 Addition and subtraction

Use + and - directly:

```
r(t) + r(t+1)
r(t) - r(t-2)
u(t) - u(t-3)
```

7.2 Amplitude scaling

Use a numeric coefficient:

```
2*r(t)
-3*u(t-2)
0.5*sin(t)
```

7.3 Multiplication and powers

The program accepts ordinary multiplication with * and converts ^ to Python's ** notation. It also understands common shorthand such as 2r(t). Using * is recommended for clarity.

```
2*r(t)
sin(2*t)*u(t)
t^2
(t-2)**2
```

8. Time shifting, scaling and reversal

These transformations are central to Signals & Systems. The program lets you place the transformation directly inside the function argument.

Input	Interpretation
$x(t-a)$	Delay / shift right by a .
$x(t+a)$	Advance / shift left by a .
$x(-t)$	Time reversal.
$x(at)$, $ a >1$	Compression in time.
$x(at)$, $0< a <1$	Expansion in time.
$Ax(t)$	Amplitude scaling by A .

8.1 Examples

```
r(t-2) -> ramp begins at t=2
u(t+1) -> step begins at t=-1
r(-t) -> reversed ramp
2*r(t) -> double amplitude
r(2*t) -> compressed time dependence
```

9. Impulse handling

A Dirac impulse $\delta(t)$ is not an ordinary finite-valued sampled function. The application therefore treats impulse terms specially: the numerical evaluator returns a zero placeholder, while the plotting layer detects simple delta terms and draws vertical arrows at the correct location.

Rule: $\delta(t-a)$ is drawn at $t=a$.

```
delta(t)
delta(t-2)
2*delta(t+1)
delta(t)+2*delta(t-3)
r(t)+2*delta(t-2)
```

10. Building signals from basic functions

The most useful educational feature is combining shifted steps and ramps to reproduce piecewise waveforms.

10.1 Rectangular pulse from steps

Turn the signal on at $t=a$ and off at $t=b$:

$$u(t-a) - u(t-b)$$

Example: $u(t)-u(t-3)$ creates a rectangular pulse from 0 to 3.

10.2 Finite ramp

Use two ramps:

$$r(t) - r(t-2)$$

Before $t=0$ it is zero, between 0 and 2 it rises linearly, and after $t=2$ the second ramp cancels the increasing slope.

10.3 Triangle from ramps

The standard triangular construction in the program is:

$$r(t) - 2*r(t-1) + r(t-2)$$

At $t=0$ the first ramp starts. At $t=1$ the second term subtracts two units of slope, changing the slope from +1 to -1. At $t=2$ the third ramp adds one unit of slope, returning the slope to zero.

10.4 Staircase

Add delayed steps:

$$u(t) + u(t-2) + u(t-4)$$

This creates successive level changes at $t=0, 2$ and 4 .

11. Analysis menu and study tools

Tool	What it reports / does
Signal Statistics	Samples, minimum, maximum, mean, RMS, numerical area and peak location.
Inspect Sample Values	Opens a table of representative t and $x(t)$ samples.
Compare with Another Signal	Overlays the current signal with a second expression.
Approximate Peak	Finds the sampled point with largest $ x(t) $.
Approximate Area	Computes numerical area over the visible time range.
Moving Value Cursor	Shows the approximate t and $x(t)$ near the mouse position.

11.1 Statistics

Statistics are computed from the sampled waveform over the current time range. This means they are numerical estimates, not symbolic results. If you change t_{min}/t_{max} , the calculated area and statistics can change.

11.2 Comparison

Enter another expression, such as $r(t-1)$, and the program overlays the second signal with the current signal using a dashed line. This is useful for visualizing time shifts and transformations.

11.3 Moving cursor

Enable Moving Value Cursor and move the mouse across the graph. The cursor snaps to the nearest sampled point and reports an approximate t and $x(t)$. Press Escape to remove it.

12. View controls

Control	Effect
Autoscale Y-axis	Fits the vertical range to the waveform and impulse arrows.
Grid	Shows or hides the graph grid.
Zero axes	Shows or hides the $t=0$ and $x=0$ axes.
Legend	Shows or hides the expression legend.
Zoom In	Halves the visible time-window width around its centre.
Zoom Out	Doubles the visible time-window width around its centre.
Fit / Reset View	Restores the default -10 to 10 time view and autoscale.
Dark Mode	Switches the interface and graph to the dark theme.

12.1 Autoscale vs manual range

Autoscale controls the Y-axis, while t_{min} and t_{max} control the horizontal range. You can therefore zoom in on time while still letting the program automatically choose a useful vertical range.

12.2 Responsive graph sizing

The graph is embedded in a Tkinter frame whose dimensions are controlled by the window. When the window is resized, the figure follows the available area. The sidebar has its own vertical scroll mechanism so long content does not need to force the whole application wider.

13. Help menu

Item	Purpose
Open Help Book	Opens this PDF from the same directory as the Python file.
Examples	Shows a separate window containing signal construction examples.
Keyboard Shortcuts	Shows the available keyboard shortcuts.
About	Shows a short description of the application.

Important: Help Book location

The program looks for the PDF relative to __file__, so the expected structure is:

```
Signals and System/  
just_trying_to_study.py  
Signals_and_Systems_Help_Book.pdf
```

If the PDF is elsewhere, Help -> Open Help Book will report that it could not find the file.

14. Saving and exporting

14.1 Save as image

Use File -> Save as Image... and choose PNG, JPG or SVG. PNG is convenient for documents; SVG is useful when a scalable vector image is desired.

14.2 Save as PDF

Use File -> Save as PDF... to export the current Matplotlib figure as a PDF.

14.3 Export CSV

Use File -> Export Samples as CSV... when you need the numerical sampled representation. The CSV contains two columns: t and $x(t)$.

14.4 Save / open expressions

Expressions can be saved as plain text and reopened later. This is convenient when maintaining a collection of solved signal examples.

15. Keyboard shortcuts

Shortcut	Action
Enter	Plot the expression when focus is in the expression field.
Cmd + Enter	Plot expression.
Cmd + N	New / clear.
Cmd + O	Open expression.
Cmd + S	Save expression.
Shift + Cmd + S	Save image.
Shift + Cmd + P	Save PDF.
Cmd + Q	Exit.
Escape	Clear the moving value cursor.

16. Worked examples

Example A - Rectangle from steps

Expression:

$$u(t) - u(t-3)$$

Interpretation: the first step turns the signal on at $t=0$. The second step subtracts one unit from $t=3$ onward, so the signal returns to zero after $t=3$.

Example B - Triangle from ramps

Expression:

$$r(t) - 2r(t-1) + r(t-2)$$

Slope analysis: before 0, slope is 0. From 0 to 1, slope is +1. At $t=1$, the $-2r(t-1)$ term changes the slope to -1. At $t=2$, the $+r(t-2)$ term changes the slope back to 0.

Example C - Shifted impulse

Expression:

$$3\delta(t+2)$$

Because $\delta(t+2) = \delta(t-(-2))$, the impulse is located at $t=-2$ and has amplitude 3.

Example D - Mixed signal

Expression:

$$r(t) + 2u(t-1) + 3\delta(t-3)$$

The graph contains a ramp, a step increase beginning at $t=1$, and an impulse at $t=3$. This is a useful exercise for learning how different building blocks coexist in one signal.

Example E - Causal damped sinusoid

Expression:

$$\exp(-0.2t) \cos(2t) u(t)$$

The exponential provides decay, $\cos(2t)$ provides oscillation, and $u(t)$ makes the signal causal.

17. Troubleshooting

The window opens but controls seem crowded

Resize the window or use the sidebar scrollbar. The toolbar has been split into compact rows and the sidebar is designed to scroll vertically rather than force the graph off-screen.

Trackpad scrolling

On macOS, move the pointer over the sidebar and use a two-finger gesture. The program routes high-resolution mouse-wheel events to the sidebar canvas. Entry fields retain their normal editing behaviour so scrolling does not interfere with text entry.

Expression error

Start with a simple expression such as $u(t)$, $r(t)$, or $\sin(t)$. Then add terms gradually. Use $*$ for multiplication when an expression becomes ambiguous:

```
Preferred: 2*r(t) - 3*u(t-2)
Also supported: 2r(t) - 3u(t-2)
```

Help Book cannot be opened

Make sure `Signals_and_Systems_Help_Book.pdf` is in the same folder as `just_trying_to_study.py`. The application resolves that location automatically using the Python file's own directory.

Tkinter import error

If Python reports `ModuleNotFoundError: No module named '_tkinter'`, the Python installation needs its matching Tk package. For the Homebrew Python 3.14 setup used in this project, the relevant package is `python-tk@3.14`.

Last study tip

Use the program actively: predict the shape on paper, type the expression, plot it, and then compare your prediction to the result. The most useful expressions for the current Signals and Systems study flow are step/ramp constructions, time shifts, scaling, reversal, and mixed step-ramp-impulse signals.